Does the J-space trade off against written chain of thought as problems get harder?

Homework Zero — Fall 2026. Model: Qwen3-4B. Code: https://github.com/avchadha/AI-Alignment.

1. Hypothesis

Recorded before the main experiments (`report/hypothesis.md`, committed 2026-07-16 before any runs; see git history): **ablating the J-space impairs the multi-hop reasoning needed to solve mathematics problems; the impairment grows with problem difficulty; and explicit chain of thought (CoT) counteracts it, with the rescue increasing with the amount of reasoning externalized.** Rationale: the paper reads the J-space and written CoT as partially interchangeable scratchpads; if the J-space is the load-bearing internal half, multi-hop work should fail without it unless externalized, and harder problems — needing more intermediate state per step — should lean on it more. Operationalized predictions: (1) direct answers suffer more from J-space ablation than CoT answers, and far more than from a matched-norm random ablation; (2) the control–ablated gap shrinks monotonically with the enforced thinking budget; (3) relative loss grows with difficulty (GSM8K → MATH-500 → AIME, and within MATH-500 levels 1–5); (4) ablated models write longer CoT when permitted.

2. Experiment design

Three ablation arms crossed with five thinking budgets on fixed problem sets (paired comparisons). **Arms:** control (no intervention); J-space ablation — at every token position, across the workspace layer band, the residual stream's projection onto the span of the k = 10 most strongly activated J-lens vectors is removed, never ablating tokens in the top-10 of a clean forward pass (the paper's procedure); random matched-norm — a projection onto k fixed random directions in the same band, rescaled per position to remove exactly the norm the J-space ablation would remove there. The random arm distinguishes J-space-specific damage from an equally large generic perturbation. **External-CoT axis:** thinking budgets {0, 256, 512, 1024, 4096} tokens in Qwen3's native thinking mode, enforced by the harness rather than by instructions: generations exceeding the budget receive a force-injected </think>, and in every arm the answer segment is teacher-forced to `The final answer is \boxed{…}` (48-token cap), because a pilot showed that instruction-based "answer directly" conditions leak CoT into the answer text. **Datasets:** GSM8K (`openai/gsm8k` test, n = 150, seeded), MATH-500 (`HuggingFaceH4/MATH-500`, n = 150, stratified 30 per level 1–5), **AIME 2024** — `HuggingFaceH4/aime_2024`, the 30 problems of the 2024 AIME I and II exams, 3 samples per problem — and a synthetic 1/2/3-hop chained-arithmetic probe (60 seeded problems), added after calibration revealed a floor effect on direct answering (§3). Each opposite outcome is visible by construction: no rescue leaves the ablated curves flat across budgets; generic damage makes the random arm track the J-space arm; a reversed difficulty interaction shows relative loss falling. **Metrics:** accuracy with 95% bootstrap confidence intervals over problems (AIME samples averaged within problem), paired McNemar tests, retained accuracy, CoT length, and budget-clip and extraction-failure rates.

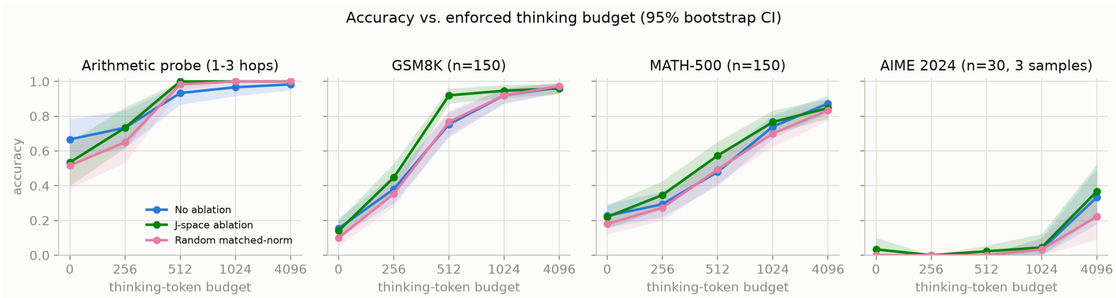
3. Experimental details

The intervention uses a pre-fitted community Jacobian lens for Qwen3-4B (`neuronpedia/jacobian-lens`, fitted on wikitext with the `anthropics/jacobian-lens` recipe); its readout reproduced the paper's three-regime walkthrough example before acceptance. The official repository provides readout code only, so the ablation is implemented here (`src/jspace/ablation.py`): J-lens vectors are computed with the final RMSNorm folded into the unembedding, the joint projection onto the selected k-vector span is removed by QR decomposition, and the clean-pass top-10 exclusion runs a second, hook-free forward pass (≈2× cost). Vectors are ranked by raw lens logit; cosine ranking was also calibrated and agreed, but promoted low-norm junk tokens. On 30 held-out calibration problems, control direct-answer accuracy is 16.7% versus 86.7% with 1,024 thinking tokens; the medium layer band (14–29 of 36) was frozen for producing the largest direct-answer drop (to 6.7%; tied with the heavy band under logit ranking) while SST-2 classification remained intact under every band and selection rule (89–90%, versus 89% control). The J-space-versus-random comparison was deliberately excluded from calibration criteria. The main run comprises 6,750 generations (3 arms × 5 budgets × 450 items) on one

H100 (≈13.5 GPU-hours), bf16, fixed seeds, resumable JSONL; answers are judged by balanced-brace `\boxed{}` extraction with `math-verify` equivalence for MATH-500. Reproduction commands and settings: `README.md` and `configs/frozen.yaml`.

4. Experimental results

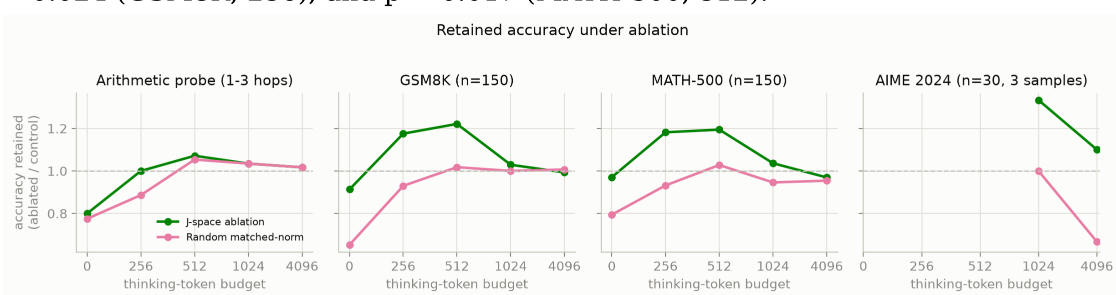
All 6,750 generations completed with no extraction failures. Full tables: `results/analysis_main.txt` and `results/extra_stats.txt`; all five figures: `report/figs/`.



budget	GSM8K ctrl	GSM8K J-abl	GSM8K rand	MATH ctrl	MATH J-abl	MATH rand
0	.153	.140	.100	.227	.220	.180
256	.380	**.447**	.353	.293	**.347**	.273
512	.753	**.920**	.767	.480	**.573**	.493
1024	.920	.947	.920	.740	.767	.700
4096	.967	.960	.973	.873	.847	.833

(a) No selective impairment at budget 0 (prediction 1 not supported). On GSM8K and MATH-500 the direct-answer baseline sits at the floor calibration predicted, and J-ablation moves it by roughly one point (McNemar $p = 0.80$ / 1.00). On the arithmetic probe, whose baseline has headroom (66.7%), J-ablation does impair direct answering (53.3%; $p = 0.0078$) with a clean hop gradient — 1-hop unaffected (100%), 2-hop 75% → 55%, 3-hop 25% → 5% — but the random arm lands in the same place (51.7%; 100/50/5% by hops; J-space versus random $p = 1.0$). The damage is multi-hop-specific but not J-space-specific.

(b) A J-space-specific accuracy gain at intermediate budgets. At budget 512 the J-ablated arm exceeds control by 16.7 points on GSM8K (92.0% vs 75.3%; McNemar 27 vs 2, $p < 10^{-5}$) and 9.3 points on MATH-500 (57.3% vs 48.0%; $p = 0.0043$), with the same sign at budget 256. Unlike the impairment in (a), this effect is specific to the J-space directions: the random arm tracks control (76.7% / 49.3% at 512), and paired J-space-versus-random tests give $p < 10^{-4}$ (GSM8K, 512), $p = 0.024$ (GSM8K, 256), and $p = 0.017$ (MATH-500, 512).



(c) The mechanism is shorter chain of thought (prediction 4 inverted). At the effectively unconstrained 4,096 budget, J-ablated CoT is consistently shorter: 1,202 versus 1,703 mean thinking tokens on GSM8K (random: 1,642), 853 versus 1,049 on the arithmetic probe, 2,326 versus 2,657 on MATH-500 (length distributions: `report/figs/fig5_cot_length.png`). The J-ablated arm is correspondingly clipped less wherever clipping is common (GSM8K at 1024: 41% versus 69%), and the gain in (b) is concentrated exactly where control chains are truncated mid-reasoning.

(d) Dose-response and convergence (prediction 2, weak form). Every arm rises steeply with budget and the arms converge by budgets 1024–4096 (GSM8K 96–97%; MATH-500 83–87%). CoT fully rescues the arithmetic impairment — the J-ablated arm reaches 100% at budget 512 (control

93.3%, random 98.3%; all arms ≥96.7% by 1024) — but from damage that was never J-space-specific.

(e) No difficulty interaction in the predicted direction (prediction 3 not supported). Within MATH-500 at budget 512 the J-ablation advantage appears at levels 1–4 (+.10, +.17, +.10, +.10) and vanishes at level 5 (.23 both; per-level curves: `report/figs/fig3_difficulty.png`). AIME is floored below budget 4096 (0–4%) and shows no significant arm differences at 4096 (control 33.3%, J-ablated 36.7%, random 22.2%); 89–94% of AIME chains still clip at 4,096 tokens, so that condition largely measures truncation.

5. Analysis of results

Of the four pre-registered predictions, only the weak form of prediction 2 (rescue by CoT, with dose–response) survives. Prediction 1's selective impairment appears only where the direct-answer baseline is off the floor, and is matched by an equally large random perturbation; prediction 3's difficulty interaction is absent; prediction 4 inverted. The one robustly J-space-specific effect is opposite in direction to the hypothesis: removing the top-k J-lens directions makes written CoT shorter (median paired length ratio 0.76) at equal or better accuracy. A post-hoc transcript probe (`scripts/verify_reflection.py`; 20 GSM8K problems regenerated with full think-text, which the main run did not store) shows the omitted content is disproportionately reflective: J-ablated chains carry fewer reflection markers per thinking token (6.7 vs 9.9 per 1,000; "wait" 1.9 vs 3.2, "alternatively" 1.6 vs 3.9, "verify" 0.04 vs 0.25) — roughly half the double-checking and re-derivation loops per problem. This suggests the ablated directions in this band carry self-monitoring or verbalization content rather than load-bearing intermediate state; on this model the J-space–CoT relationship resembles modulation of how much the model narrates, not an internal scratchpad for which text substitutes. Two qualifications: floor effects — when externalization is genuinely prevented, un-ablated Qwen3-4B scores 15–23% on GSM8K/MATH-500 and 0% on AIME, leaving little internal reasoning to ablate, a caution against extrapolating the paper's headline result to small thinking-tuned models — and alternative explanations: the wikitext-fitted community lens may capture a shallower subspace than a paper-faithful lens, the random arm matches removed norm per position but not its location in the residual basis, and thinking-mode sampling adds variance. None of these plausibly explains the intermediate-budget gain, which replicates across two datasets with the random arm flat. With more time or compute: a paper-faithful lens fit; k and layer-band sweeps with per-position delta-norm audits; clamping or patching individual J-vectors to isolate the CoT-shortening direction; faithfulness checks on the shortened chains; and replication on a larger model whose direct-answer baseline is off the floor without a synthetic probe.